The challenge

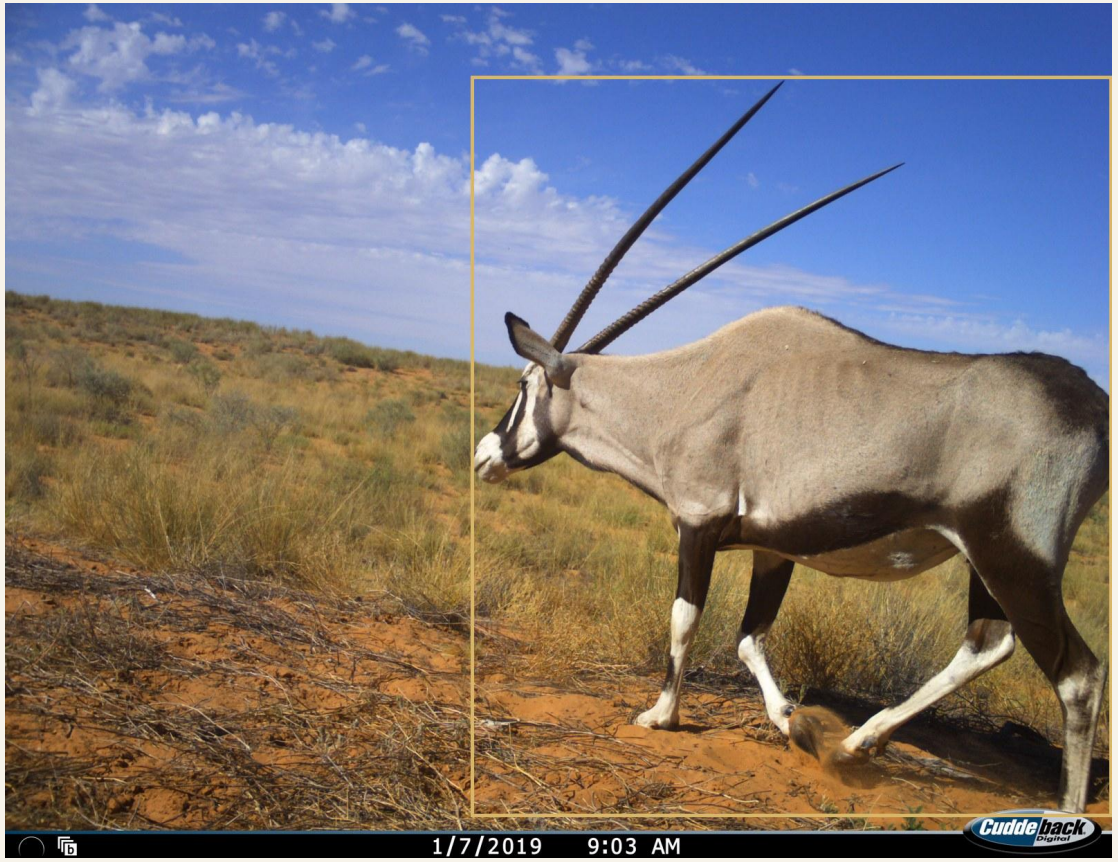
KAMouflage Vision

How might we enable MOCCAE environmental monitoring teams, conservation authorities, and researchers to identify local wildlife species and analyze patterns of their observed presence and spatial-temporal activity across monitored habitats, to reduce manual camera-trap image processing, accelerate validated observations, and improve the timeliness of conservation insights under image-quality, species-identification, spatial-coverage, and scientific-verification constraints, using camera-trap imagery, species reference records, observation timestamps, location metadata, and computer vision and artificial intelligence while aligning with the UAE's biodiversity conservation and sustainability objectives?

Khaled Sandoqa and Abdulla Ezzaldeen

Challenge wording reproduced in the project notes

Our solution: KAMouflage Vision



Detect

Find animals in a batch of camera-trap images.

Suggest

Identify a species and combine burst evidence.

Verify

A reviewer confirms or corrects the event.

Understand

Confirmed events feed the map and activity chart.

Real inference example: gemsbok. Public southern-African proxy image. Scores remain uncalibrated.

A focused fit for the MOCCAE pilot

Wildlife Insights	AddaxAI	KAMouflage Vision
Cloud platform for camera-trap management, AI identification and human validation.	Open-source desktop tool. Models run locally, with offline operation after setup.	Local prototype with UAE government reference data, logged corrections and confirmed observations.

Our proposed advantage

A workflow tailored to MOCCAE, with government data integration and a path to authority-hosted deployment.

Deployment fit is the hypothesis. No head-to-head accuracy or unique-feature superiority has been demonstrated.

Sources: wildlifeinsights.org/about-wildlife-insights-ai; wildlifeinsights.org/get-started/identify-images; docs.addaxai.com/docs/help/faq/

Evidence for UAE conservation decisions

86.8%

Animal bursts retained

165 / 190

67.4%

Empty bursts filtered

58 / 86

48.9%

Source labels matched

93 / 190

Verified records can guide conservation decisions.

Supporting species monitoring and habitat protection aligns our proposed workflow with UAE biodiversity and sustainability priorities.

UAE context: 44 usable reserve points and 381 bird-list rows from MOCCAE via Bayanat, plus GBIF species references.

600 photos / 276 bursts. Exploratory southern-African proxies; simulated UAE cameras. Same-data iteration and unknown pretrained overlap. Review remains mandatory; UAE impact and time savings are unmeasured. Policy: UAE Environment Policy (u.ae).

Business Model Canvas

How we deliver	The value	How adoption works
Partners Proposed authority sponsor, reserve team and ecologists.	**Value** Less repetitive review and traceable observations.	**Customers** Monitoring staff use it. Conservation authorities sponsor it.
Activities Inference, verification, evaluation and reporting.	**Relationships** Reviewer onboarding, support and quality reviews.	**Channels** A challenge-owner pilot and government integration.
Resources Permitted images, expert labels, software and hosting.	**Costs** Engineering, review labor, compute, storage and security.	**Funding** Sponsored pilot, then scoped support or integration contracts.

Proposed business model. Buyer demand, partnerships and pricing still require validation.

The next 90 days

01-30

Establish the baseline

Prepare local GPU, database and storage. Agree access controls and a local test set.

31-60

Improve local quality

Fine-tune SpeciesNet on verified UAE crops. Calibrate and test held-out sites.

61-90

Test real operations

Review new retrievals. Audit errors and compare human time. Make a scale decision.

The ask: one site, permitted images and expert reviewers.

Proposed targets: 99% animal retention, 95% accepted precision and 50% less review time.

Existing-hardware resource scenario: AED 43,176, including labor and contingency. New GPU/server purchases need separate quotes.

Government images, training, inference, databases and backups stay on authority-controlled infrastructure.

The MOCCAE wildlife monitoring challenge

KAMouflage Vision

How might we enable MOCCAE environmental monitoring teams, conservation authorities, and researchers to identify local wildlife species and analyze patterns of their observed presence and spatial-temporal activity across monitored habitats, to reduce manual camera-trap image processing, accelerate validated observations, and improve the timeliness of conservation insights under image-quality, species-identification, spatial-coverage, and scientific-verification constraints, using camera-trap imagery, species reference records, observation timestamps, location metadata, and computer vision and artificial intelligence while aligning with the UAE's biodiversity conservation and sustainability objectives?

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Challenge wording reproduced in the project notes

Technical evidence and deployment boundary

Backup / What works today and what needs pilot development

Implemented MVP	Funded pilot work
Real MIT detector and Apache SpeciesNet classifier. Actual species names and a documented footer exclusion for proxy images. Full detection sidecar and resumable caches.	Local taxonomy coverage, calibrated thresholds and independent tests on previously unseen UAE sites.
Append-only review decisions. Reviewer-confirmed analytics. Restricted image placeholders. Prototype CSV export.	Authenticated access, protected-location policies, monitoring, backup/retention controls and validated Camtrap DP packages.
Local processing and cached demo. No cloud inference API required.	Authority-approved UAE deployment, security review, operational ownership and support agreements.

Licenses: THIRD_PARTY_NOTICES.md. Production hosting and data reuse require authority approval.

Evaluation protocol and limitations

Backup / Reproducibility and scientific interpretation

Unit of evaluation	276 complete source bursts. Source labels are sequence-level. No frame-level mAP or animal-count claim.
Exact species scoring	Case-insensitive source-name aliases. Any actual label can match in a multi-label burst. Higher-taxon outputs count as incorrect for exact species.
Data limitations	Curated class balance, small rare-class samples and unknown overlap with pretrained training. Exploratory iteration on the same dataset. No independent UAE generalization claim.
Risk control	All species require review. No automatic retraining. No-detection candidates remain reviewable, because missed animals can occur.
Next experiment	A site-separated local test set and an assisted-vs-manual time study with monitoring staff.

Local deployment and training budget

Resource	Planning assumption	AED
Integration engineering	160 hours x AED 100	16,000
Expert labels and review	40 hours x AED 150	6,000
Model adaptation	40 hours x AED 100	4,000
Local GPU allocation	60 hours x assumed AED 8 internal cost	480
Database, storage and backup	90-day internal allocation allowance	1,500
Access and security hardening	40 hours x AED 100	4,000
Support and onboarding	40 hours x AED 100	4,000
Contingency and total	20% of AED 35,980 = AED 7,196	43,176

Assumes an existing authority GPU and server environment. Excludes new hardware procurement. Capacity, retention, security scope and actual rates require IT sizing and quotes.

Likely judge questions

Backup / Short answers grounded in the MVP

Did you train a new model?	We integrated pretrained models and implemented the workflow. Local adaptation is a pilot task.
Why these models?	A general animal detector separates detection from species recognition. Their selected artifacts have permissive licenses.
Is it tested in the UAE?	The current evaluation uses southern-African proxies. UAE field accuracy needs permitted local data.
How is this different?	We propose authority-specific verification, open government data integration and local operational support. Customer validation is still required.
What happens when it is wrong?	The reviewer corrects the event. Decisions persist, and corrections become candidate labels only after quality checks.

Sources and judging coverage

Backup / Pioneers working rubric from the supplied playbook, pages 7-8

Criterion	Weight	Evidence in the main pitch
Challenge alignment	15%	Problem, review workflow and pilot outcome
Data	15%	Bayanat reserve/bird data, GBIF references and proxy disclosure
AI	15%	Real inference image, measured errors and review policy
Feasibility	20%	90-day gates, dependencies and resource budget
Market potential	15%	Buyer, adoption route, support model and validation task
Pitch	20%	Three-minute story, team participation and canvas page

Sources: supplied HackArena Playbook, Q&A; and Abu Dhabi agenda; MOCCAE/FCSC via Bayanat: Protected Lands Coordinates in UAE and List of local Aves (Birds) in the UAE 2023; GBIF; LILA Snapshot Karoo/Kgalagadi; microsoft/Pytorch-Wildlife; google/cameratrapai.

Reference URLs: lila.science/datasets/snapshot-kgalagadi • github.com/google/cameratrapai • microsoft.github.io/MegaDetector/model_zoo/ • gbif.org/terms • camtrap-dp.tdwg.org

The Word rubric screenshot totals 110%. We use the playbook’s 100% Pioneers table pending organizer clarification. The requested presentation limit is three minutes plus two minutes of questions.

Local AI training and government data

Area	Proposed funded pilot design
Training model	Fine-tune SpeciesNet's EfficientNetV2-M classifier on expert-verified UAE animal crops. Start with the classification head, then selected backbone layers if justified.
Detection	Retain MDV6-mit-yolov9-c. Fine-tune the detector only if local recall requires it and bounding-box labels are available.
Local infrastructure	Authority-controlled GPU compute, PostgreSQL/PostGIS, image/object storage and encrypted backups. Training artifacts and labels remain inside the approved boundary.
Validation and security	Site-separated test data, model versioning and signed-off promotion. Role-based access, audit logs, encryption and key custody require authority security approval.
Cloud boundary	Default: on-premises. Any UAE private-cloud alternative requires written authority approval. Public Colab/Kaggle is for public proxy data only, never confidential government archives.

MVP uses pretrained artifacts only. No local fine-tuning or production security certification is claimed today. Architecture source: github.com/google/cameratrapai/blob/main/model_cards/v4.0.1a.md